



School: Campus:

Academic Year: Subject Name: Subject Code:

Semester: Program: Branch: Specialization:

Date:

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiment : Build a Use Case – Tokenized Supply Chain

* Coding Phase: Pseudo Code / Flow Chart / Algorithm

1. Identify Participants:

Define all stakeholders — Manufacturer, Supplier, Transporter, Distributor, Retailer, and Customer.

2. Token Creation:

Create digital tokens representing goods or product batches. Each token contains product ID, origin, and value.

3. Smart Contract Setup:

Establish smart contracts that govern token transfer, payment release, and verification of product conditions.

4. Product Registration:

The manufacturer registers the product by assigning tokens that represent the items being shipped.

5. Token Transfer Through Supply Chain:

As the product moves through different stages, ownership of tokens is transferred from one participant to the next (e.g., Manufacturer → Supplier → Distributor → Retailer → Customer).

6. Verification and Validation:

Each transfer is verified by the blockchain network to ensure authenticity and prevent double-spending of tokens.

7. Automatic Settlement:

Once the product reaches the customer, smart contracts automatically release payment to the concerned parties.

8. Record Maintenance:

All transactions remain permanently stored on the blockchain, providing complete transparency and traceability.

Software used:

1. VS Code.
2. MS Word.
3. Brave for researching.

* Implementation Phase: Final Output (no error)

- All participants in the supply chain are enrolled as authorized members of the blockchain network.
- The manufacturer generates digital tokens that symbolize the physical goods ready for shipment.
- Upon confirmation of material receipt, the tokens are securely transferred to the supplier.
- As the goods progress through the logistics chain, token ownership is sequentially handed over to the transporter, distributor, and retailer.
- Every transfer and transaction is permanently logged and verified on the blockchain ledger.
- Smart contracts automatically release payments once predefined delivery conditions are satisfied.
- The end consumer obtains both the physical product and its corresponding digital proof of ownership (token).
- The blockchain ledger maintains a transparent, tamper-proof record of the entire product journey from manufacturer to customer.

* Observations:

Blockchain technology enables **tokenization**, which simplifies the tracking of assets and guarantees the authenticity of products throughout the supply chain.

Through **smart contracts**, business operations become automated, reducing the need for manual intervention and ensuring that all transactions follow pre-set trust-based conditions.

These contracts execute automatically once the required terms are fulfilled.

The system provides **real-time transparency**, allowing every participant to view and verify data instantly.

Such visibility strengthens accountability and promotes trust among all stakeholders.

Digital tokens of ownership represent the goods securely, protecting against duplication and counterfeit products.

This ensures that every product's origin and journey can be verified at any time.

Additionally, **automated payment mechanisms** allow faster settlements, improving efficiency in operations.

Financial transactions occur seamlessly once delivery confirmations are made.

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/ Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student:

Name :

Regn. No. :

Signature of the Faculty: